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Product Testing & Vulnerability Assessment Guide

A complete guide for security engineers and developers to test their applications for post-quantum cryptographic vulnerabilities using the PQ-Sentry platform.

📋 Version 1.0.0 🛡️ PQ-Sentry Platform 📅 May 2026

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01 What is PQ-Sentry and Why You Need It

PQ-Sentry is a Post-Quantum Cryptography Readiness Platform built for Kubernetes environments. It scans, inventories, and audits the cryptographic algorithms used by your microservices, then provides actionable migration paths to quantum-safe standards (NIST FIPS 203/204/205).

> ℹ️ **KEY VALUE PROPOSITION**
>
> PQ-Sentry answers one critical question: **"If a quantum computer existed today, which of my services would be immediately compromised?"** — and then gives you the tools to fix it with zero application code changes.

Core Capabilities

CAPABILITY	DESCRIPTION	TESTING FOCUS
CBOM Inventory	Cryptographic Bill of Materials — real-time catalog of all cipher suites in use	Verify accurate algorithm detection and risk classification
SBOM ↔ CBOM Bridge	Correlates software dependency CVEs with cryptographic cipher vulnerabilities	Verify unified risk scoring and CVE-to-cipher mapping
AI Risk Advisor	GenAI-powered remediation playbooks with Kubernetes-native manifests	Verify playbook accuracy and actionability
Quantum Migration	One-click cipher upgrade from classical → post-quantum (ML-KEM-768)	Verify live database migration and dashboard refresh
Compliance Audits	Auto-generated reports for SOC2, FedRAMP, NIS2, CMMC 2.0	Verify regulatory checklist accuracy

02 The Quantum Threat — Understanding the Risk

Cryptographically Relevant Quantum Computers (CRQCs) threaten all public-key cryptography based on integer factorization (RSA) and discrete logarithms (ECDSA). Using **Shor's Algorithm**, a CRQC can break these in polynomial time.

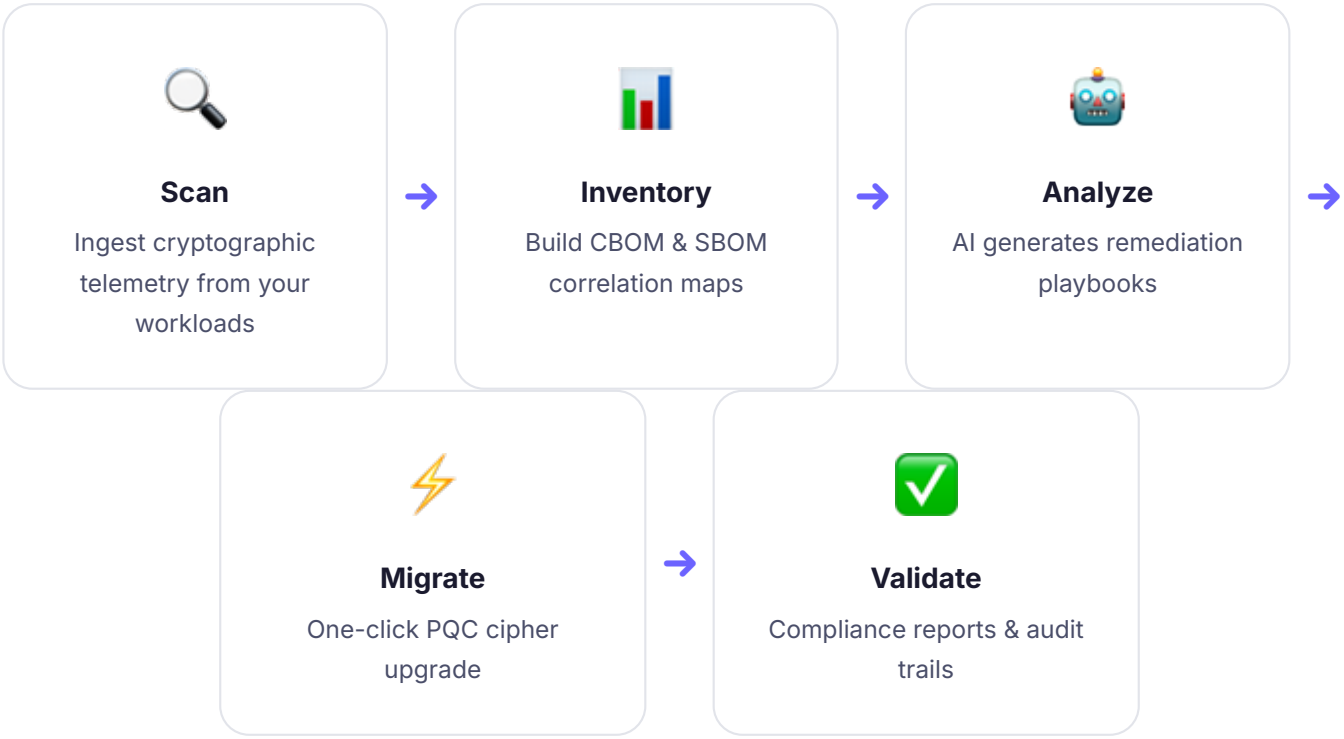
Vulnerability Classification

ALGORITHM	TYPE	KEY SIZE	QUANTUM RISK	NIST REPLACEMENT
RSA-2048	Asymmetric (Factoring)	2048	CRITICAL	ML-KEM-768 (FIPS 203)
RSA-4096	Asymmetric (Factoring)	4096	CRITICAL	ML-KEM-1024 (FIPS 203)
ECDSA P-256	Elliptic Curve (DLOG)	256	HIGH	ML-DSA-65 (FIPS 204)
ECDSA P-384	Elliptic Curve (DLOG)	384	HIGH	ML-DSA-87 (FIPS 204)
AES-256-GCM	Symmetric	256	SAFE	No change (Grover: $\sqrt{}$ complexity)
ML-KEM-768	Lattice-based (PQC)	768	PQC SAFE	Already quantum-resistant
ML-DSA-65	Lattice-based (PQC)	n/a	PQC SAFE	Already quantum-resistant
None / Plaintext	Unencrypted	0	CRITICAL	Immediate TLS required

> ⚠️ **HARVEST NOW, DECRYPT LATER (HNDL)**
>
> Adversaries are **already recording encrypted traffic today** to decrypt it once quantum computers are available. Services handling PII, financial transactions, health records, or classified data are at immediate risk — even before CRQCs exist.

03 How PQ-Sentry Testing Works

The testing workflow follows a 5-stage pipeline. Each stage produces verifiable outputs:



04 Setting Up the Testing Environment

> ✅ **PREREQUISITES**
>
> You only need **Docker Desktop** (v24.x+). No Java, Maven, PostgreSQL, or build tools required.

STEP 1

Extract the Distribution Package

```
# Unzip the package you received
unzip pq-sentry-dist.zip
cd pq-sentry-dist
```

STEP 2

Launch PQ-Sentry

```
chmod +x start.sh stop.sh
./start.sh
```

Wait ~30-60 seconds until you see: 🎉 `PQ-SENTRY IS READY FOR TESTING`

STEP 3

Open the Dashboard

Open `dashboard/index.html` in Chrome, Firefox, or Edge. The dashboard connects to the local API on port 8080.

```
# macOS
open dashboard/index.html

# Linux
xdg-open dashboard/index.html
```

STEP 4

Verify API Health

```
curl http://localhost:8080/api/actuator/health
# Expected: {"status":"UP"}
```

05 Test Scenario A — Dashboard Vulnerability Discovery

Objective: Verify that PQ-Sentry correctly identifies and classifies quantum-vulnerable microservices.

ACTION

Open the Dashboard Overview

Navigate to the Overview page (loads by default). Inspect the CBOM table listing all microservices.

Expected Results — Pre-Seeded Demo Data

SERVICE	NAMESPACE	ALGORITHM	KEY SIZE	RISK LEVEL	QUANTUM SAFE?
old-rsa	payments	RSA-2048	2048	CRITICAL	❌ No
auth-service	auth	ECDSA-P256	256	HIGH	❌ No
new-pqc	payments	ML-KEM-768	768	SAFE	✅ Yes

Validation Checklist

- [ ] Risk Score ring chart displays a score ≥ 70 (CRITICAL range)
- [ ] Total Services count = 3
- [ ] Critical count = 1 (old-rsa)
- [ ] High count = 1 (auth-service)
- [ ] Safe count = 1 (new-pqc)
- [ ] Each row has color-coded risk badges matching the table above
- [ ] Clicking the ✨ Consult button redirects to the AI Advisor

06 Test Scenario B — SBOM ↔ CBOM Correlation Audit

Objective: Verify that PQ-Sentry correctly correlates software package vulnerabilities (SBOM) with cryptographic posture (CBOM).

ACTION

Navigate to SBOM ↔ CBOM Bridge

Click **"SBOM ↔ CBOM Bridge"** in the left sidebar. The split-view panel loads correlated data.

Expected CVE Correlations

SERVICE	PACKAGE	VERSION	CVE	SEVERITY	CIPHER RISK	UNI SC(
old-rsa	openssl	1.1.1	CVE-2023-0286	CRITICAL	CRITICAL	100
old-rsa	openssl	1.1.1	CVE-2023-3817	HIGH	CRITICAL	100
auth-service	boringssl	fips-2021	CVE-2022-0778	HIGH	HIGH	85
new-pqc	oqs-openssl	3.1.0-pqc	*None*	CLEAN	SAFE	10

Validation Checklist

- [ ] Split panel shows CBOM on left, SBOM on right
- [ ] CVE badges display with correct severity colors
- [ ] Unified risk score combines both SBOM + CBOM factors
- [ ] `new-pqc` shows clean status with no CVEs
- [ ] Correlation insights text is displayed for each workload

07 Test Scenario C — AI-Powered Remediation Playbook

Objective: Verify the AI Advisor generates accurate, actionable remediation plans for vulnerable services.

ACTION

Consult the AI Advisor

Click ✨ **Consult** next to `old-rsa` in the CBOM table (or navigate to AI Advisor in the sidebar). Select the service, then click **Consult AI Advisor**.

Expected Playbook Contents

- [ ] **Vulnerability Summary** — Explains Shor's Algorithm risk for RSA-2048
- [ ] **NIST Compliance Target** — References FIPS 203 (ML-KEM-768) and FIPS 204 (ML-DSA-65)

- [ ] **YAML Manifest** — Contains a valid `QuantumReadinessPolicy` custom resource with:
  - Correct namespace matching the service
  - `algorithm: ML-KEM-768`
  - `signatureAlgo: ML-DSA-65`
  - Canary rollout strategy
- [ ] **kubectl Commands** — Executable deployment commands provided
- [ ] **Rendering** — Markdown renders with proper code blocks, headings, and formatting

08 Test Scenario D — Live Quantum Migration ⭐

> ⭐ CORE FEATURE TEST
>
> This is the **primary test**. The migration engine actually updates all classical ciphers in the database to post-quantum standards and triggers a complete E2E dashboard refresh.

STEP 1

Open the Policies Page

Click **Policies** in the left sidebar. You'll see a YAML editor and an operator terminal panel.

STEP 2

Configure the Migration

Uncheck the "Dry-Run Mode" checkbox. This enables the live migration mode.

> 💡 **DRY-RUN VS. LIVE**
>
> **Dry-Run ON** = Simulates the migration without making changes. **Dry-Run OFF** = Actually migrates ciphers in the database.

STEP 3

Execute the Migration

Click ▶ **Apply Policy**. Watch the operator terminal.

Expected Terminal Output

```
[INFO]  Parsing QuantumReadinessPolicy: 'production-pqc'
[INFO]  Rollout strategy configured: Canary
[INFO]  Target namespaces: default, payments, auth, billing
[INFO]  Scanning cbom_entries for quantum-vulnerable workloads...
[INFO]  Upgrading old-rsa from RSA-2048 → ML-KEM-768 (PQC Safe)
[INFO]  Injecting PQC Envoy sidecar into old-rsa pods...
[INFO]  Upgrading auth-service from ECDSA-P256 → ML-KEM-768 (PQC Safe)
[INFO]  Injecting PQC Envoy sidecar into auth-service pods...
[INFO]  Recalculating unified cluster risk score...
[SUCCESS] 2 services upgraded to post-quantum cipher suites.
[SUCCESS] Cluster risk score updated: CRITICAL → SAFE (score: 0)
```

Expected Dashboard Changes (Auto-Refresh)

METRIC	BEFORE MIGRATION	AFTER MIGRATION
Risk Score	≥ 70 (CRITICAL)	0 (SAFE)
Critical Services	1	0
High Services	1	0
Safe Services	1	3
old-rsa Algorithm	RSA-2048	ML-KEM-768
auth-service Algorithm	ECDSA-P256	ML-KEM-768
Compliance Status	WARN / PARTIAL	COMPLIANT

Validation Checklist

- [ ] Terminal shows step-by-step log animation (typewriter effect)
- [ ] All 3 services now display ML-KEM-768 with SAFE badges
- [ ] Risk score ring chart updates to 0
- [ ] Overview metrics auto-refresh without manual reload
- [ ] SBOM ↔ CBOM Bridge reflects updated cipher suites
- [ ] Compliance audit indicators turn green
- [ ] Running migration again reports "No quantum-vulnerable workloads detected"

09

Test Scenario E — Compliance Audit Report Generation

Objective: Verify that the platform generates accurate compliance reports for multiple regulatory frameworks.

ACTION

Generate and Export the Report

On the Overview page, click **Export PDF**. A new browser tab opens with the compliance report. Use `Ctrl+P` / `Cmd+P` to save as PDF.

Frameworks Covered

FRAMEWORK	CONTROLS CHECKED	PRE-MIGRATION STATUS	POST-MIGRATION STATUS
SOC2 TSC	CC6.1, CC6.7	WARN	PASS
FedRAMP	PQC Transition, FIPS 140-3	FAIL / PARTIAL	PASS
EU NIS2	Art 21.1, Art 21.2	WARN	PASS
CMMC 2.0	SC.L2-3.13.11	WARN	PASS

Validation Checklist

- [ ] Report opens in new tab with styled, print-ready layout
- [ ] All four compliance frameworks are displayed
- [ ] Detailed CBOM inventory table is included
- [ ] Print/PDF output renders cleanly (white background, no cutoffs)

10 Test Scenario F — Ingesting Your Own Application Data

> 🔧 **ADVANCED TESTING**
>
> You can ingest your own application's cryptographic data to see how PQ-Sentry would classify and handle your real services.

Use the `POST /api/cbom/{clusterId}/ingest` endpoint to push custom CBOM telemetry. This simulates what the PQ-Sentry CLI scanner would report from a live Kubernetes cluster.

How to Audit Your Own Application

STEP 1

Identify Your Cryptographic Posture

Before ingesting, gather this information about each of your services:

FIELD	DESCRIPTION	EXAMPLE VALUES
`serviceName`	Name of your microservice	`payment-gateway` , `user-api`
`namespace`	Kubernetes namespace	`production` , `staging`
`algorithm`	TLS/cipher algorithm in use	`RSA-2048` , `ECDSA-P256` , `AES-256`
`keySize`	Key size in bits	`2048` , `256` , `384`
`tlsEnabled`	Whether TLS is active	`true` / `false`
`certExpiry`	Certificate expiration (ISO 8601)	`2026-12-31T00:00:00Z`

STEP 2

Reset the Environment (Optional)

```
docker compose down -v
./start.sh
```

STEP 3

Ingest Your Service Data

```
curl -X POST http://localhost:8080/api/cbom/00000000-0000-0000-0000-00
  -H "Content-Type: application/json" \
  -d '[
    {
      "serviceName": "payment-gateway",
      "namespace": "production",
      "algorithm": "RSA-2048",
      "keySize": 2048,
      "tlsEnabled": true,
      "certExpiry": "2026-12-31T00:00:00Z"
    },
    {
      "serviceName": "user-api",
      "namespace": "production",
      "algorithm": "ECDSA-P256",
      "keySize": 256,
      "tlsEnabled": true,
      "certExpiry": "2027-03-15T00:00:00Z"
    },
    {
      "serviceName": "internal-cache",
      "namespace": "infrastructure",
      "algorithm": "None",
      "keySize": 0,
      "tlsEnabled": false,
      "certExpiry": null
    }
  ]'
```

STEP 4

View Results in Dashboard

Refresh `dashboard/index.html`. Your custom services now appear alongside the demo data. You can:

- See your services' risk classifications
- Run the AI Advisor against your specific services
- Apply the quantum migration to upgrade them to ML-KEM-768
- Generate compliance reports including your services

> 💡 **TIP — TEST DIFFERENT SCENARIOS**
>
> Try ingesting services with various algorithms (`RSA-4096`, `DES-56`, `3DES`, `AES-128`) to see how PQ-Sentry classifies and prioritizes different cryptographic risk levels.

11 API Reference for Custom Testing

Base URL: `http://localhost:8080/api` | Default Cluster ID: `00000000-0000-0000-0000-000000000001`

METHOD	ENDPOINT	DESCRIPTION	AUTH
GET	/actuator/health	API health check	None
GET	/cbom/{clusterId}	List all CBOM entries	None
GET	/cbom/{clusterId}/summary	Aggregated risk summary	None
POST	/cbom/{clusterId}/ingest	Ingest custom CBOM telemetry	API Key*
GET	/sbom/{clusterId}/bridge	SBOM ↔ CBOM correlation	None
POST	/ai/advise	Generate AI remediation playbook	None
POST	/policies/{clusterId}/apply	Apply quantum migration	None
GET	/reports/compliance/{clusterId}	Generate compliance report	None

* API Key auth is enforced in production. In the sandbox testing environment, ingestion is open for ease of testing.

12 Vulnerability Classification Matrix

Use this matrix to understand how PQ-Sentry classifies the algorithms found in your applications:

RISK LEVEL	SCORE RANGE	ALGORITHMS	ACTION REQUIRED	TIMELINE
CRITICAL	90 - 100	RSA-2048, RSA-1024, DES, 3DES, MD5, SHA-1, None/Plaintext	Immediate migration to ML-KEM-768	Now
HIGH	60 - 89	ECDSA P-256, ECDH, DSA, RSA-3072	Schedule migration within 6 months	6 months
MEDIUM	30 - 59	RSA-4096, ECDSA P-384, X25519 (hybrid candidate)	Plan migration, monitor NIST updates	12 months
SAFE	0 - 29	ML-KEM-768, ML-DSA-65, SLH-DSA, AES-256, SHA-3	No action needed	Compliant ✅

13 Test Acceptance Criteria Checklist

Use this master checklist to sign off on your testing session:

Functional Tests

- [ ] **F-001:** Dashboard loads and displays 3 pre-seeded services with correct risk levels
- [ ] **F-002:** Risk score ring chart renders with correct value and color
- [ ] **F-003:** SBOM ↔ CBOM Bridge displays correlated CVE-to-cipher data
- [ ] **F-004:** AI Advisor generates a well-structured remediation playbook
- [ ] **F-005:** Policy migration (live mode) upgrades 2 vulnerable services to ML-KEM-768

- [ ] **F-006:** Dashboard auto-refreshes post-migration without manual page reload
- [ ] **F-007:** Compliance report opens in new tab with correct framework evaluations
- [ ] **F-008:** All API endpoints return valid JSON responses

Post-Migration Verification

- [ ] **M-001:** Risk score = 0 after migration
- [ ] **M-002:** All services show ML-KEM-768 / SAFE
- [ ] **M-003:** Compliance badges change from WARN to PASS
- [ ] **M-004:** Re-running migration correctly reports "No vulnerable workloads"

Performance & UX

- [ ] **P-001:** API health check responds within 500ms
- [ ] **P-002:** CBOM data loads within 2 seconds on page open
- [ ] **P-003:** Terminal log animation plays smoothly (no stuttering)
- [ ] **P-004:** All interactive elements have hover/click feedback

Data Ingestion (Optional)

- [ ] **D-001:** Custom CBOM entries appear in dashboard after ingestion
- [ ] **D-002:** Custom services can be migrated to post-quantum ciphers
- [ ] **D-003:** Database reset (`docker compose down -v`) clears all data

14 Resetting the Environment

Soft Reset (Keep data, restart services)

```
./stop.sh
./start.sh
```

Hard Reset (Delete all data, start fresh)

```
docker compose down -v
./start.sh
```

> 💡 **WHEN TO HARD RESET**
>
> After testing the quantum migration, all services will already be upgraded. To re-test the migration flow, perform a hard reset to re-seed the original vulnerable data.

15 FAQ & Troubleshooting

Q

Do I need Java or Maven installed?

No. Everything runs inside Docker containers. You only need Docker Desktop.

Q

Can I test with my real Kubernetes cluster data?

Yes. Use Section 10 (Scenario F) to ingest your own service data via the API. In the future, the PQ-Sentry CLI scanner will automate this by scanning live cluster TLS configurations.

Q

The dashboard shows "Failed to fetch" errors

The backend API isn't running or hasn't finished starting. Run `curl http://localhost:8080/api/actuator/health` to check. If it fails, run `docker logs pq-sentry-api` to see errors.

Q

Port 8080 is already in use

Stop the other application using port 8080, or check what's using it: `lsof -i :8080` (macOS/Linux).

Q

Can I run the migration more than once?

Yes. If all services are already migrated, it will report "No quantum-vulnerable workloads detected." To re-test, perform a hard reset: `docker compose down -v && ./start.sh`

Q

Is this testing against a real quantum computer?

No. PQ-Sentry identifies cryptographic algorithms that would be vulnerable to quantum attacks and helps you migrate proactively, before quantum computers are available.

PQ-Sentry — Post-Quantum Cryptography Readiness Platform

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